

Module – 1(Basics of Computer graphics and Algorithms)

Basics of Computer Graphics and its applications. Video Display devices- Refresh Cathode Ray Tubes, Random Scan Displays and systems, Raster scan displays and systems. Line drawing algorithms- DDA, Bresenham's algorithm. Circle drawing algorithms- Midpoint Circle generation algorithm, Bresenham's algorithm.

Computer graphics refers to the creation, storage and manipulation of image data (pictures and drawings) using a digital computer with the help of a specialized graphical system.

- Display those images on screens or hardcopy devices
- studies methods for digitally synthesizing and manipulating visual content.
- Used in various fields to represent data.
- Scientific research, engineering applications, medicine, business, industry, government, art, entertainment, advertising, education and other fields make use of computer graphics.
- It enhances the communication between computers and users.

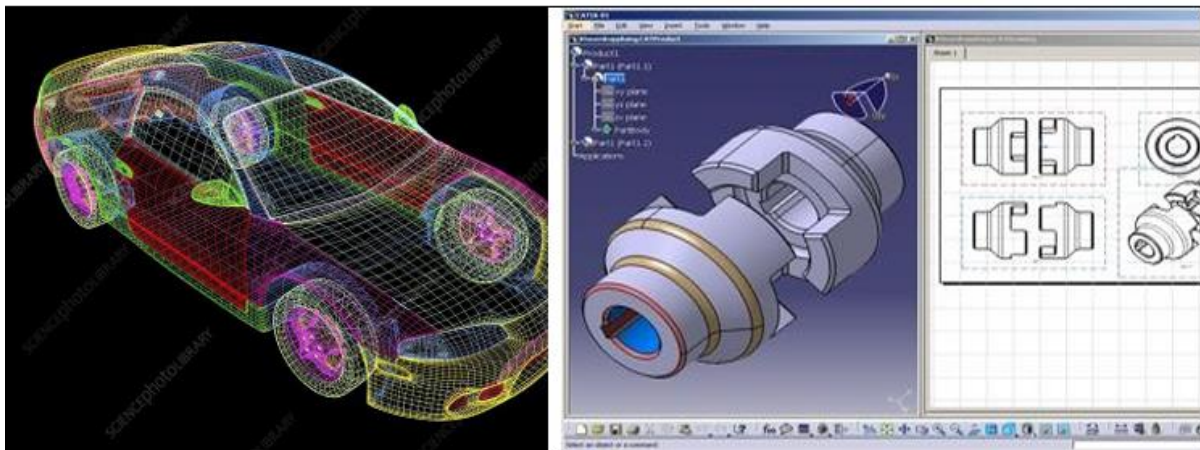
Applications of Computer Graphics

1) Computer-Aided Design

CAD methods are now routinely used in the design of buildings, automobiles, aircraft, watercraft, spacecraft computers, textiles, and many other products.

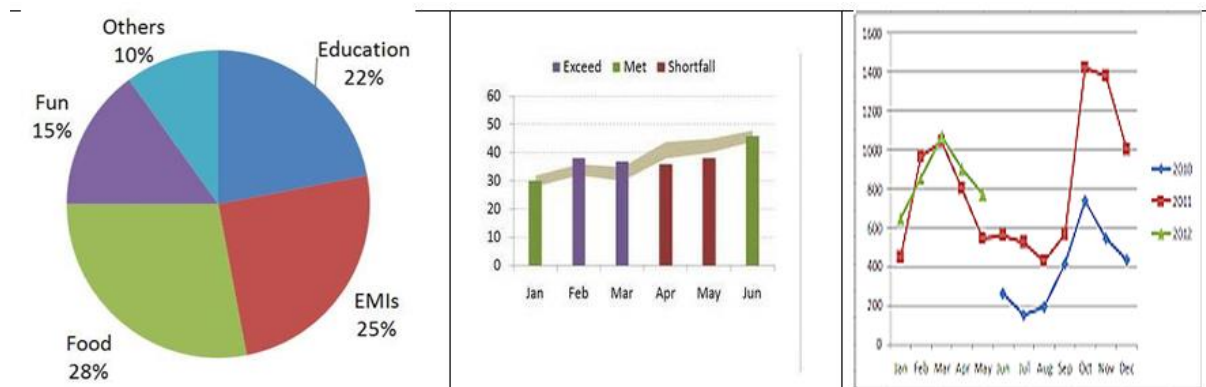
For some design applications, objects are first displayed in a wireframe outline form that shows the overall shape and internal features of objects. Wireframe displays allow designers to quickly see the effects of interactive adjustments to design shapes.

Animations are often used in CAD applications. Real-time animations using wire frame displays on a video monitor are useful for testing the performance of a vehicle or system.



2) Presentation Graphics

Presentation Graphics programs are used to produce illustrations for reports or to generate PowerPoint presentations. It is commonly used to summarize financial, statistical, mathematical, scientific, and economic data for research reports, managerial reports consumer information bulletins, and other types of reports.



3) Computer Art

Computer graphics methods are widely used in both fine art and commercial art applications.

Artist's paintbrush programs (Lumena), Pixel Paint and Super Paint, symbolic, CAD packages, desktop publishing software, animation packages that provide facilities for designing object shapes and specifying object motions. We have now Adobe illustrator, affinity designer, sketch, Corel Draw etc



4) Entertainment

Computer graphics methods are now commonly used in making motion pictures, music videos, and television shows. Sometimes graphics objects are displayed by themselves, and sometimes graphics objects are combined with the actors and live scenes.

A common graphics method employed in many commercials is morphing, where one object is transformed into another.

2. Face of Young person is converted into aged person as shown in fig:



Movie Industry

Computer Graphics is about animation (films)

Game Industry

- Focus on interactivity
- Cost effective solutions
- Avoiding computations and other tricks

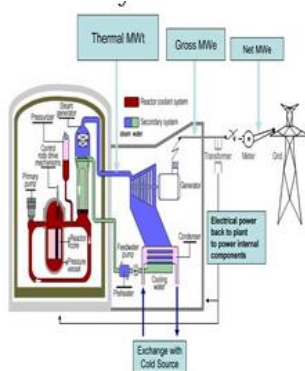
5) Education and Training

Computer-generated models of physical, financial, and economic systems are often used as educational aids.

Models of physical systems, physiological systems, population trends, or equipment, such as color-coded diagrams can help trainees to understand the operation of the system.

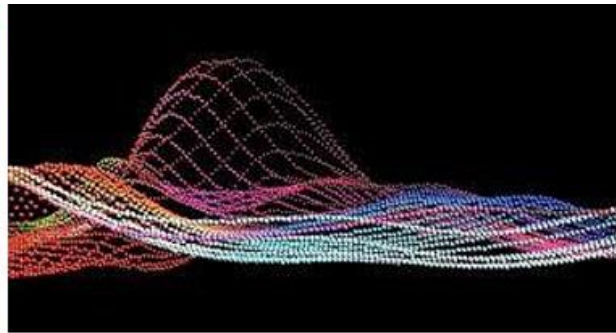
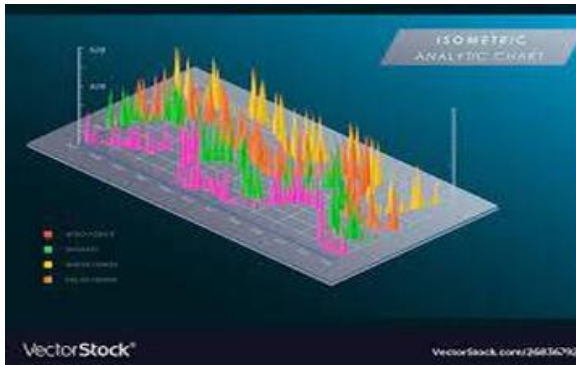
Interactive tutorials, simulations, and visual demonstrations improve learning.

Simulators for practice sessions and training: ship captains, aircraft pilots, heavy equipment operators ,air traffic-control personnel



6) Visualization

Scientific engineers, medical personnel, business analysts, and others often need to analyse large amounts of information or study the behaviour of certain processes. But if the data are converted to a visual form the trends and patterns are often immediately apparent.



7) Image Processing

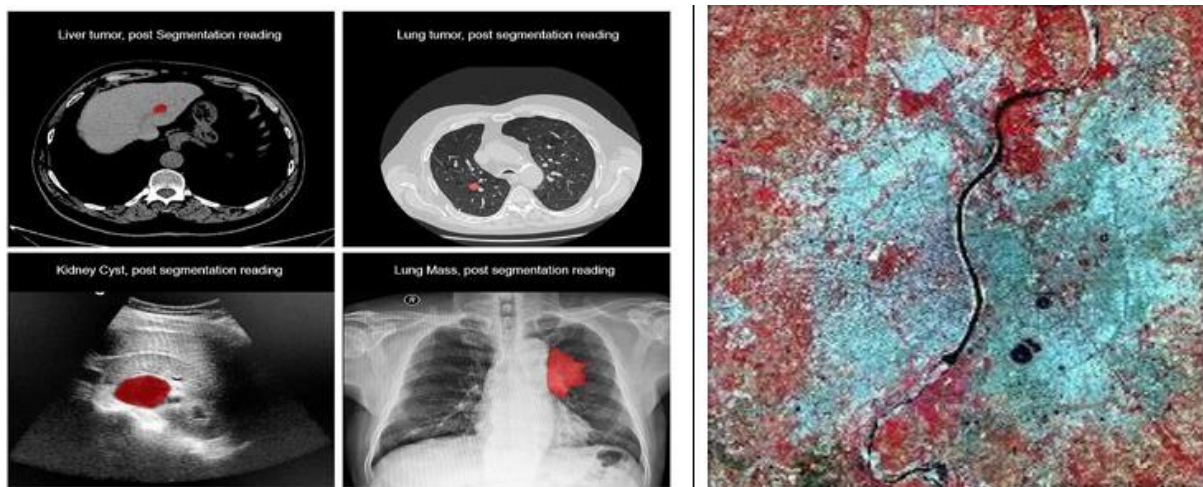
Image processing applies techniques to modify or interpret existing pictures, such as photographs and TV scans.

Medical applications-Picture enhancements, Tomography, Simulations of operations, Ultrasonics & nuclear medicine scanners

Applications of image processing-

Improving picture quality

Machine perception of visual information (Robotics)



8) Graphical User Interfaces

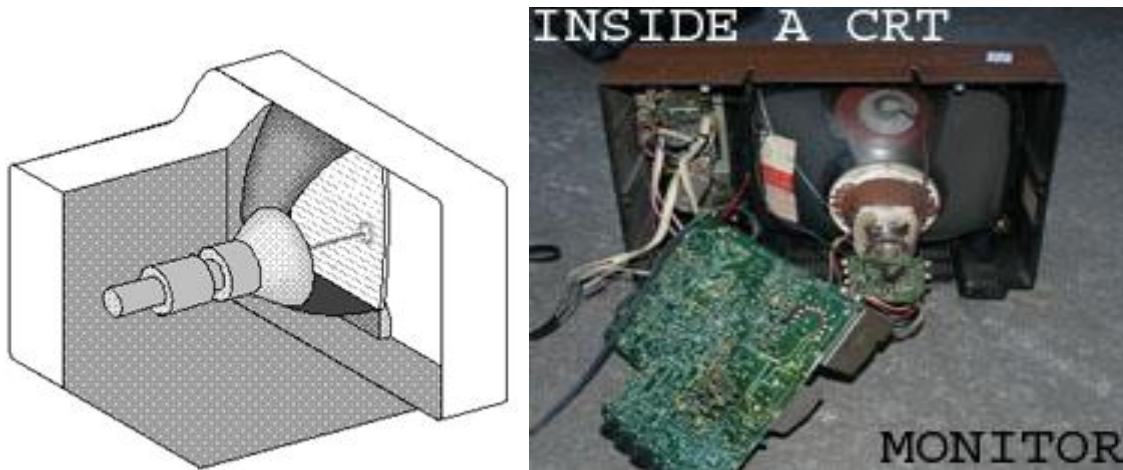
- Major component – Window manager (multiple-window areas)
- To make a particular window active, click in that window (using an interactive pointing device)
- Interfaces display – menus & icons
- Icons – graphical symbol designed to look like the processing option it represents
- Advantages of icons – less screen space, easily understood
- Menus contain lists of textual descriptions & icons

Video Display Devices

- Refresh Cathode Ray Tubes
- Random Scan Displays and systems
- Raster Scan Displays and systems

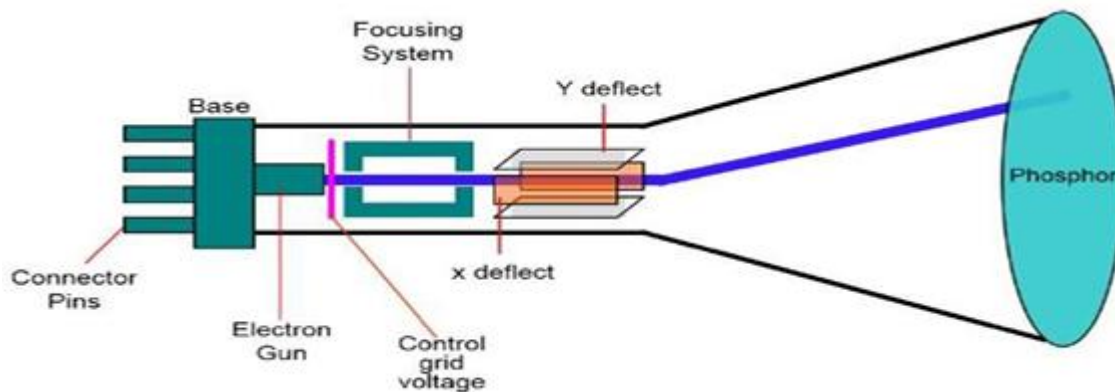
The primary output device in a graphical system is the video monitor.

The operation of most video monitors is based on the standard Cathode-Ray Tube (CRT) design.



1) Refresh Cathode Ray Tubes (CRT)

In a graphic system, the video monitor is a primary output device and the operation of most the video monitors is based on the Cathode Ray Tube.



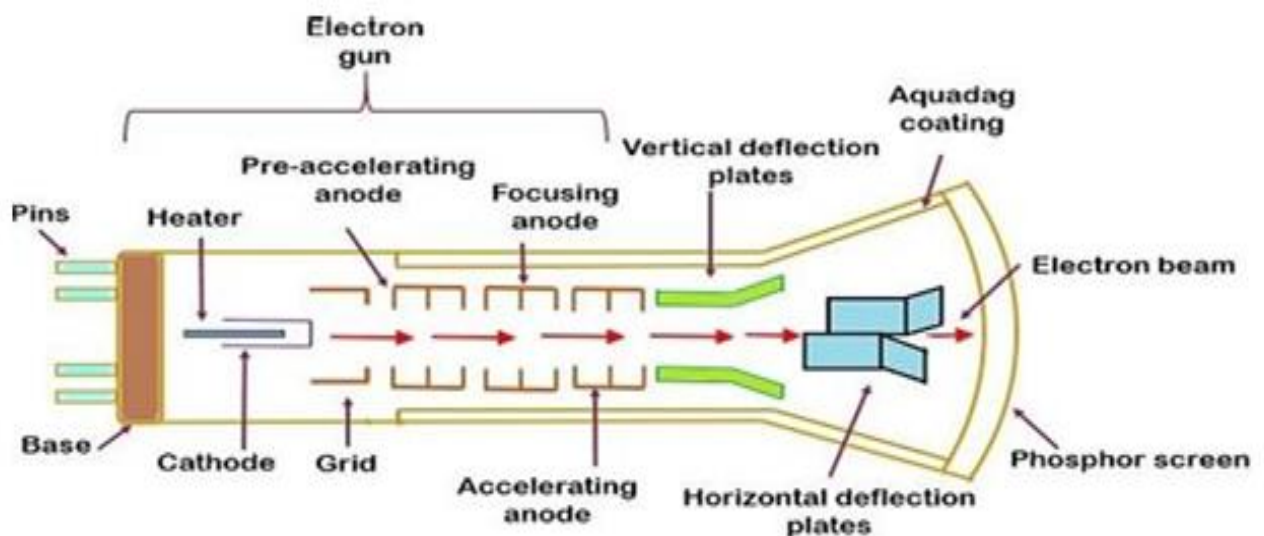
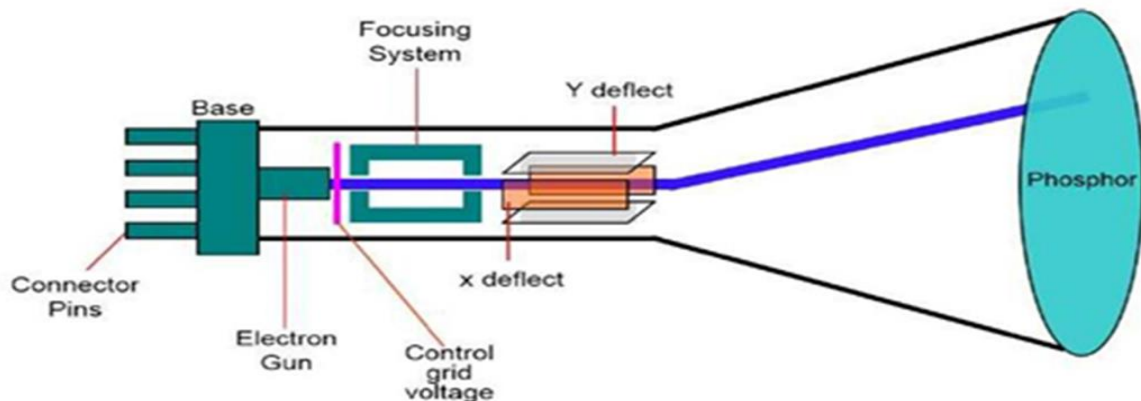
In a cathode-ray tube a beam of electrons, emitted by an electron gun, passes through focusing and deflection systems, that direct the beam towards a specific position on the **phosphor-coated screen**. The part of the phosphor screen emits light where the electron beam strikes it. The light emitted by the phosphor fades very rapidly, one way to keep the phosphor glowing is to redraw the picture repeatedly by quickly directing the electron beam back over the same points. This type of display is called a **refresh CRT**.

Computer monitors often have a maximum refresh rate. This number, measured in hertz (Hz),

determines how many times the screen is redrawn each second. Typical refresh rates for CRT monitors include 60, 75, and 85 Hz.

Components of CRT

1. Electron Gun
2. Focusing and accelerating anode
3. Vertical and horizontal deflection plates
4. Phosphorous coated screen- evacuated glass envelope with a phosphorescent screen that glows visibly when struck by the electron beam



The primary components of an electron gun in a CRT are the heated metal cathode and a control grid. Heat is supplied to the cathode by directing a current through a coil of wire, called the **filament**, inside the cylindrical cathode structure. This causes electrons to be boiled off the hot cathode surface. In the vacuum inside the CRT envelope, the free negatively charged electrons are then accelerated towards the phosphor coating by a high positive voltage.

The amount of light emitted by the phosphor coating depends on the number of electrons

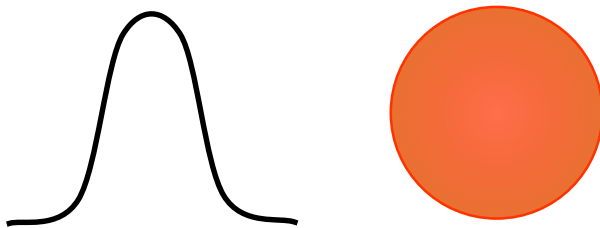
striking the screen. We specify the intensity level for individual screen positions with graphics software commands.

Phosphor Persistence

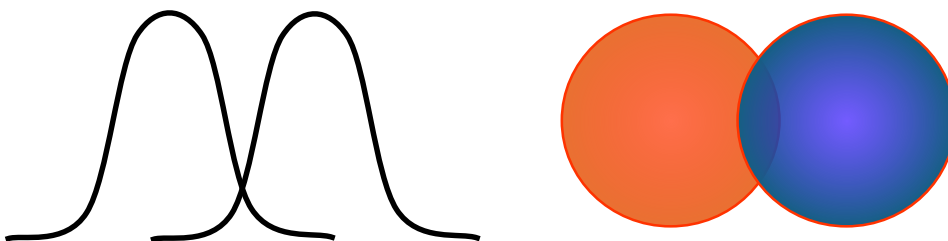
- The time it takes the emitted light from the screen to decay to one-tenth of its original intensity.
- There are different kind of phosphors for use in a CRT.
- Besides color, a major difference is their persistence – how long they continue to emit light after the CRT beam is removed.
- A phosphor with low persistence is useful in animation.
- A high-persistence phosphor is useful for displaying highly complex, static pictures.
- Graphics monitors are usually constructed with a persistence in the range from 10 to 60 microseconds.

Brightness (Intensity) distribution

- The intensity is greater at the center of the spot, and it decreases with Gaussian distribution out to the edges of the spot.



- Two illuminated phosphor spots are distinguishable when their separation is greater than the diameter at which a spot intensity has fallen to 60% of maximum (at the center of the spot).
- Spot size depends on intensity.



Note:

A computer monitor's resolution refers to the approximate number of pixels the device is capable of displaying. It's expressed as the number of horizontal dots by the number of

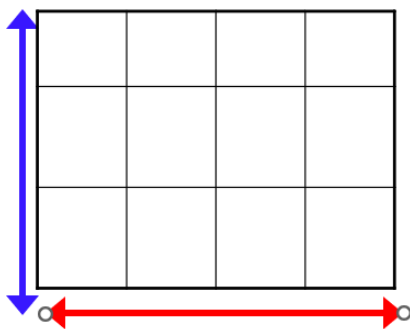
vertical dots: for example, an 800 x 600 resolution means the device can show 800 pixels across by 600 pixels down. In total, this screen displays 480,000 pixels.

Aspect Ratio

- **Aspect ratio** gives the ratio of horizontal points to vertical points necessary to produce equal-length lines in both directions on the screen.
- Aspect ratio 4:3 (most common) means that a horizontal line plotted with 4 points has the same length as a vertical line plotted with 3 points.

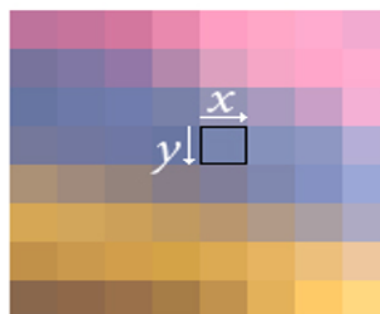
The **aspect ratio** is the ratio between the width and height of the screen.

Eg: 32:9, 21:9, 16:9 etc.



Aspect Ratio

Frame Aspect Ratio (FAR) =
horizontal/vertical size
TV 4 : 3
HDTV 16 : 9

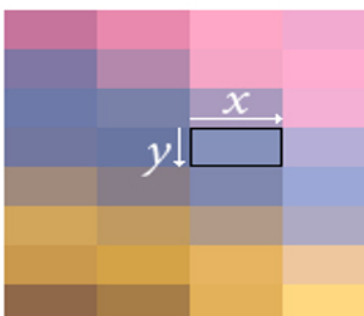


Pixel Aspect Ratio

$$\text{PAR} = x/y = 1:1$$

x = pixel width
y = pixel height

Pixel Aspect Ratio (PAR)
= pixel width/ pixel height
Nuisance in graphics if
not 1



Pixel Aspect Ratio

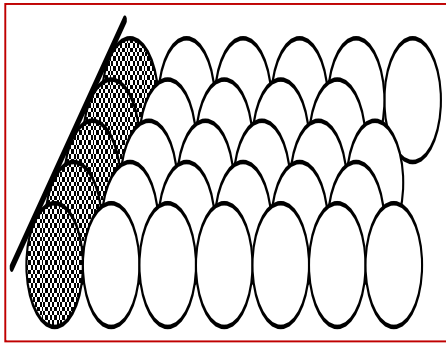
$$\text{PAR} = x/y = 2:1$$

x = pixel width
y = pixel height

Pixel

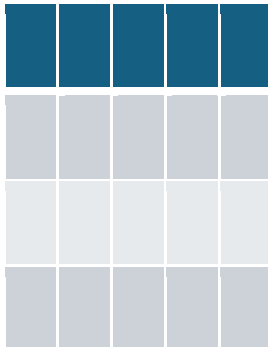
- Addressability is the number of individual dots per inch (d.p.i.) that can be created.
- The picture on a screen consists of intensified points.

- The smallest addressable point on the screen is called **pixel** or picture element.



Resolution

- **Resolution-** The maximum number of points (pixels) that can be displayed without overlap on a screen and determines the detail that can be seen in an image.
- Determines Clarity of image.



- number of points per centimeter that can be plotted horizontally and vertically (i.e. 1280×1024).
- The physical size of a graphics monitor, on the other hand, is given as the length (in inches) of the screen diagonal.
- Resolution depends on type of phosphor, intensity to be displayed, and focusing and deflection systems.

Refresh Rate

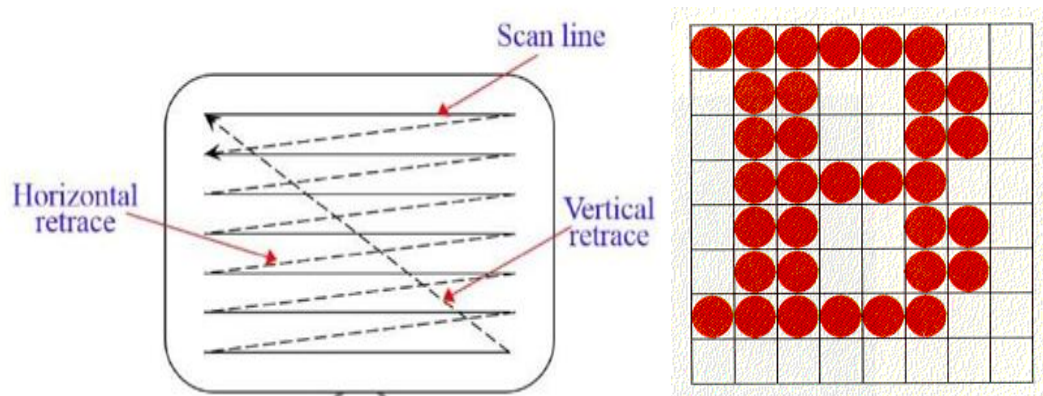
- ♦ The number of times per second the image is redrawn.
- ♦ The entire contents of the frame buffer are displayed on the CRT at a rate high enough to avoid flicker. This rate is called the refresh rate.

- ♦ For a human to see a steady image on most CRT displays, the same path must be retraced, or refreshed, by the beam at least 60 times per second.
- ♦ Current raster-scan displays perform refreshing at the rate of 60 to 80 frames per second, although some systems now have refresh rates of up to 120 frames per second.
- ♦ Refresh rates are described in units of cycles per second, or Hertz (Hz), where a cycle corresponds to one frame (i.e. a refresh rate of 60 frames per second = 60 Hz).

Refresh-rate for Films and TVs

- The display technology on a monitor is different from that of film.
- A film projector can maintain the continuous display of a film frame until the next frame is brought into view.
- On a video monitor a phosphor spot begins to decay as soon as it is illuminated.
- On films, below 24 frames per second, we can perceive a gap between successive screen images.
- Old silent films show flicker because they were photographed at a rate of 16 frames per second.
- When sound systems were developed in the 1920s, motion picture film rates increased to 24 frames per second removing flickering.
- Today TV refresh rate is 25 frames per second in Europe and 30 frames per second in the USA.

2) Raster Scan Displays



Raster Scan Displays are the most common type of graphics monitor which employs CRT.

It is based on television technology.

In a raster scan system electron beam sweep across the screen, from top to bottom covering one row at a time. A pattern of illuminated pattern of spots is created by turning beam intensity on and off as it moves across each row. A memory area called **refresh buffer or frame buffer** stores picture definition. This memory area holds intensity values for all screen points. Stored

intensity values are restored from the frame buffer and painted on-screen taking one row at a time. Each screen point is referred to as a pixel or pel (shortened form of picture element).

- A picture is produced as an array - raster of picture elements.
- These elements are called Pixels or Pels (Picture Elements).
- A pixel corresponds to a location, or small area, in the image
- Pixel - Dot on screen, Is addressable, Can be illuminated
- More pixels → higher resolution
- Dot pitch - distance between dots
- Smaller distance → better quality image
- the pixel bits are stored in a part of memory called the refresh buffer or frame buffer.

he **refresh rate** (or "vertical refresh rate", "vertical scan rate") is the number of times per second that a raster-based display device redraws images.

On cathode ray tube (CRT) displays, higher refresh rates produce less flickering, thereby reducing eye strain.

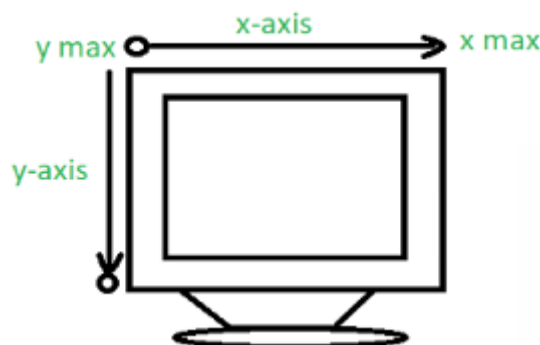
In raster scan systems refreshing is done at a rate of 60-80 frames per second.

Refresh rates are also sometimes described in units of **cycles per second / Hertz (Hz)**.

At the end of each scan line, the electron beam begins to display the next scan line after returning to the left side of the screen. The return to the left of the screen after the refresh of each scan line is known as the **horizontal retrace** of the electron beam.

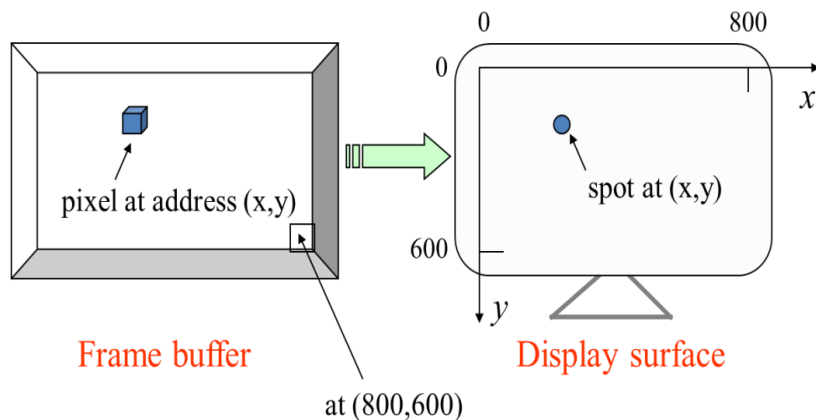
At the end of each frame, the electron beam returns to the top left corner and begins the next frame, this movement is known as **vertical retrace**.

➤ Frame buffer locations and corresponding screen positions are referenced in Cartesian coordinates.



The origin of the coordinate system for identifying screen positions is usually

specified in the lower-left corner



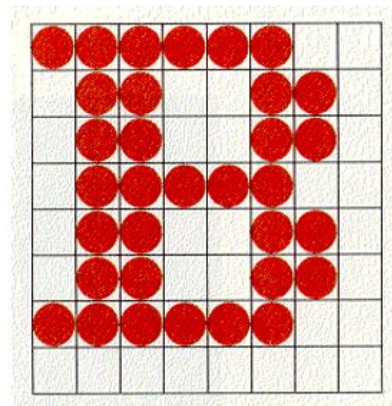
- The number of bits per pixel in the frame buffer is called **depth or bit planes**
- Determines properties such as how many colors can be represented on a given system.
- Buffer with 1 bit per pixel (1-bit-deep) – **Bitmap** allows 21 colors (black and white)
- Buffer with multiple bits per pixel (8-bit-deep) – **Pixmap** allows 28 colors (=256)
- Full color (RGB-color systems), 24 (or more) bits per pixel in order to display sufficient colors to represent most images realistically.

The Bit-map

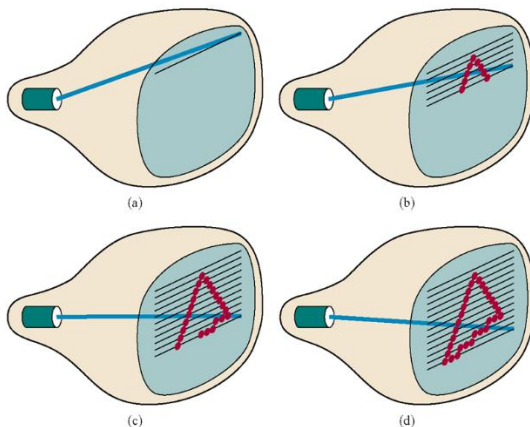
1-bit-deep frame-buffer

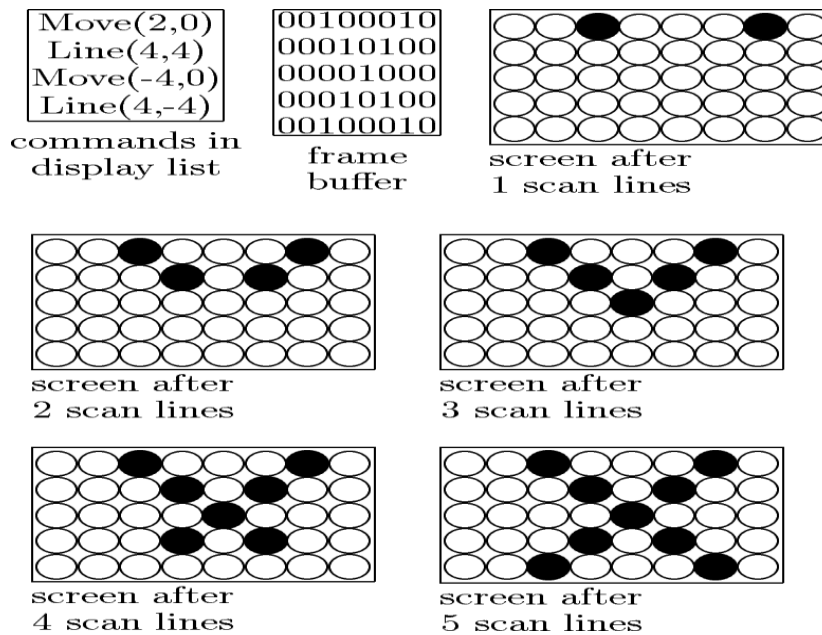
1	1	1	1	1	1	0	0
0	1	1	0	0	1	1	0
0	1	1	0	0	1	1	0
0	1	1	1	1	1	0	0
0	1	1	0	0	1	1	0
0	1	1	0	0	1	1	0
1	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0

Frame buffer



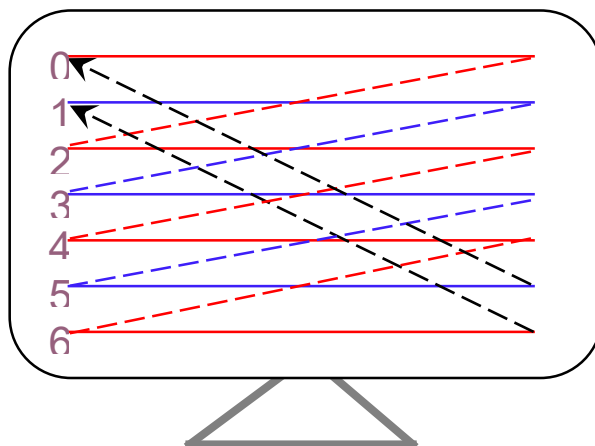
Pixels

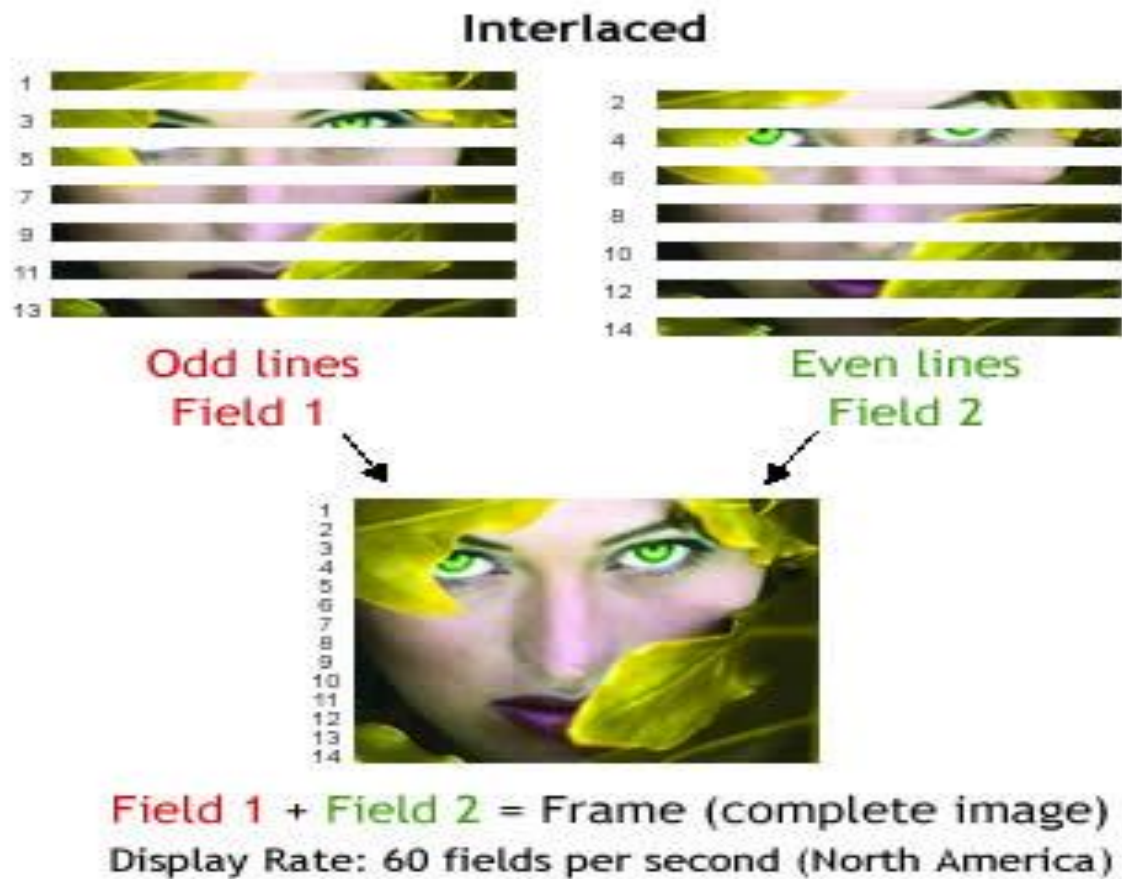




Interlaced Raster Scan Displays

- To reduce flicker, divide frame into two fields—one consisting of the even scan lines and the other of the odd scan lines.
- Even and odd fields are scanned out alternately to produce an interlaced image. Odd/even scan lines covered in successive refresh cycles.
- Hence 30 frames/sec refresh rate appears like 60 frames/sec.





Raster Scan System

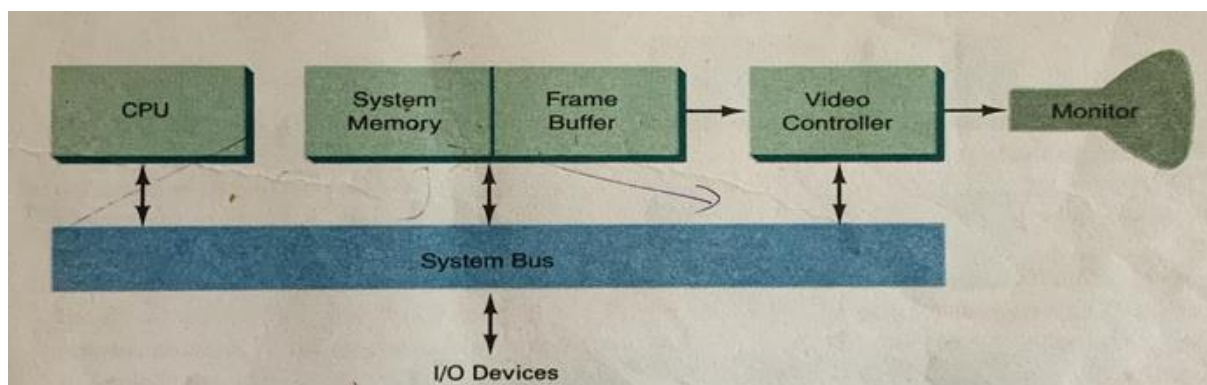
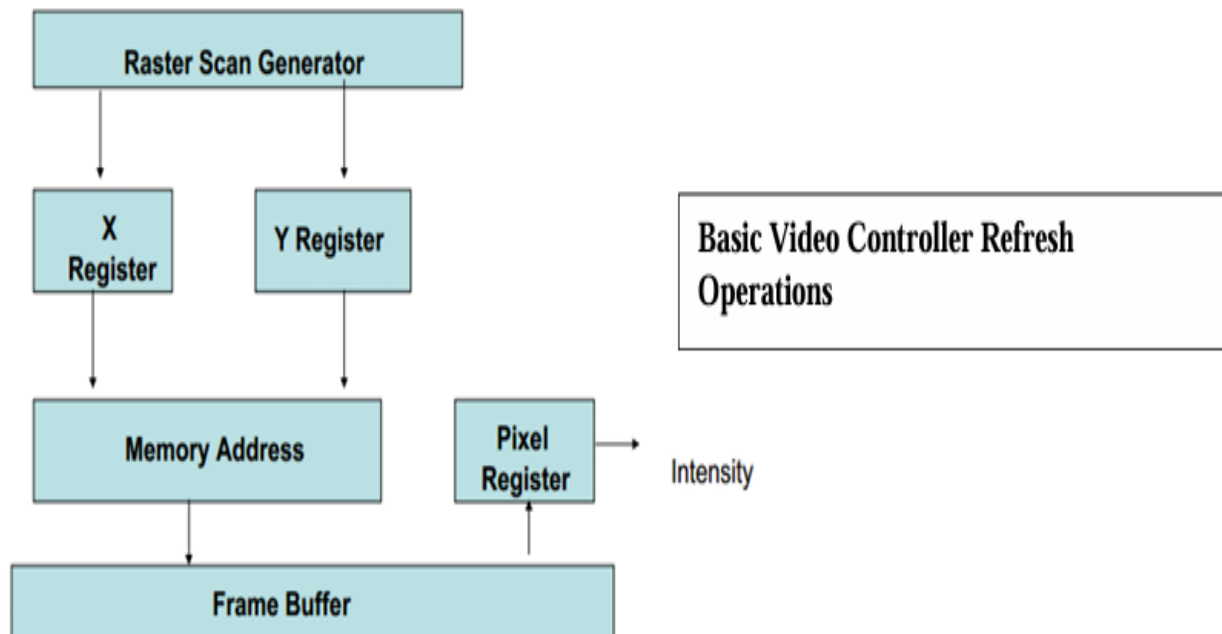


Fig: -Architecture of a simple raster graphics system

In addition to the central processing unit, a special purpose processor called the **video controller or video display controller** is used to control the operation of the display device. A fixed area of the system memory is reserved for the frame buffer, and the video controller is given direct access to the frame buffer memory.

Basic video controller refresh operations

Note: Scanning start from top left corner, so coordinate for first pixel is (xmin,ymax) .



Two registers are used to store the coordinates of the screen pixels. Initially, the x registers are used to store the coordinates of the screen pixels. Initially x register is set to 0 and y register is set to y max. The value stored in the frame buffer for this pixel position is then retrieved and used to set the intensity of the CRT beam. Then the x register is incremented by 1, and the process is repeated for the next pixel on the top scan line. This procedure is repeated for each pixel along the scan line. After the last pixel on the top scan line has been processed, the x register is reset to 0 and the y register is decremented by 1. Pixels along this scan line are then processed in turn and the procedure is repeated for each successive scan line.

After cycling through all pixels along the bottom can line ($y=0$), the video controller resets the registers to the first-pixel position on the top scan line and the refresh process starts over.

Usually, cycle time is too slow if processing pixels one after another, to speed up pixel processing, video controllers can retrieve multiple pixel values from the refresh buffer on each pass. The multiple pixel intensities are then stored in a separate register and used to control the CRT beam intensity for a group of adjacent pixels. When that group of pixels has been processed, the next block of pixel values is retrieved from the frame buffer.

Note: In high-quality systems, for example, two frame buffers are often provided so that one buffer can be used for refreshing while the other is being filled with intensity values.

Raster -Scan Display Processor

Figure shows a raster system containing a separate display processor, referred to as a **graphics controller or a display processor**. The purpose of the display processor is to free the CPU

from the graphics chores. In addition to the system memory, a separate display processor memory area can also be provided.

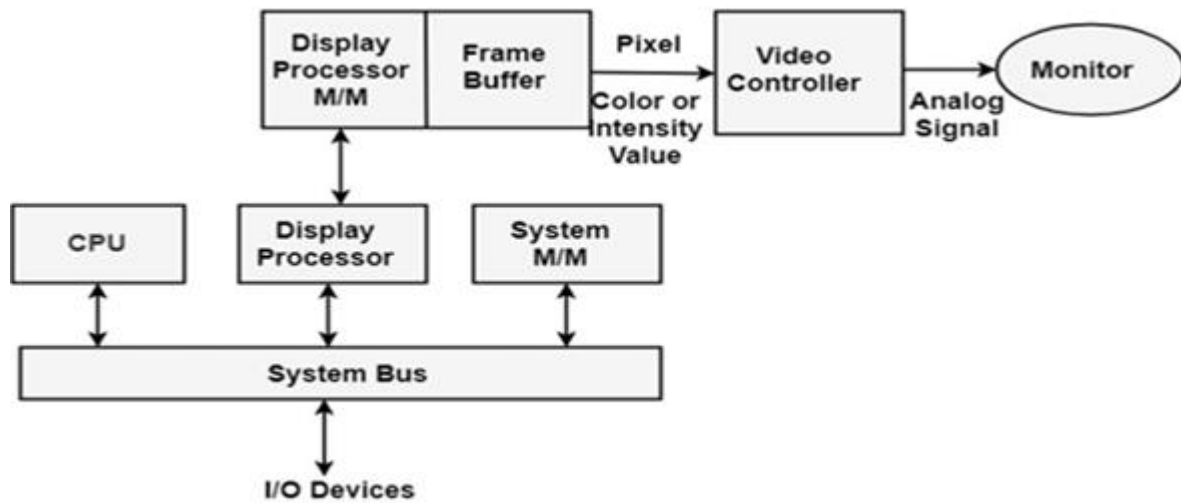


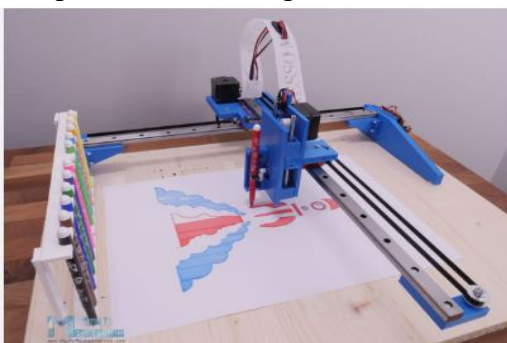
Fig:-Architecture of a raster -graphics system with a display processor.

A major task of the display processor is digitizing a picture definition given in an application program into a set of pixel intensity values for storage in the frame buffer. This digitization process is called scan conversion.

3) Random Scan Display

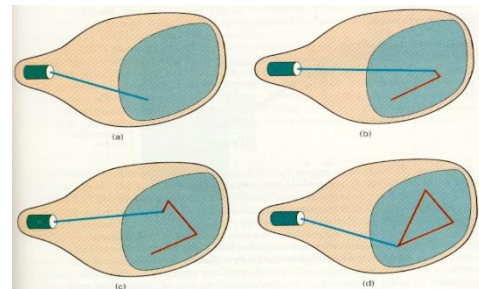
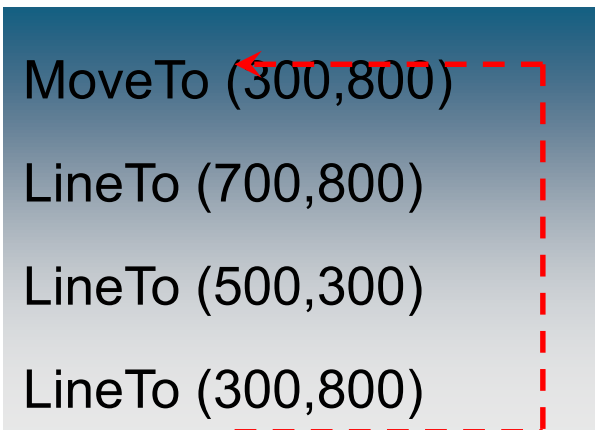
In Random-Scan Display electron beam is directed only to the areas of screen where a picture must be drawn. It is also called **vector display**, as it draws picture one line at time. It can draw and refresh component lines of a picture in any specified sequence.

- Pen plotter is an example of random-scan displays.



- Vector stands for line.
- Developed in the mid-60's and in common use until the mid- 80's
- The electron beam is directed only to parts of the screen where the picture is to be drawn.

The number of lines regulates refresh rate on random-scan displays. An area of memory called **refresh display files** stores picture definition as a set of line drawing commands. The system returns back to the first-line command in the list after all the drawing commands have been processed. High-quality vector systems can handle around 100, 00 short lines at this refresh rate. Faster refreshing can burn phosphor. To avoid this every refresh cycle is delayed preventing a refresh rate greater than 60 frames per second.

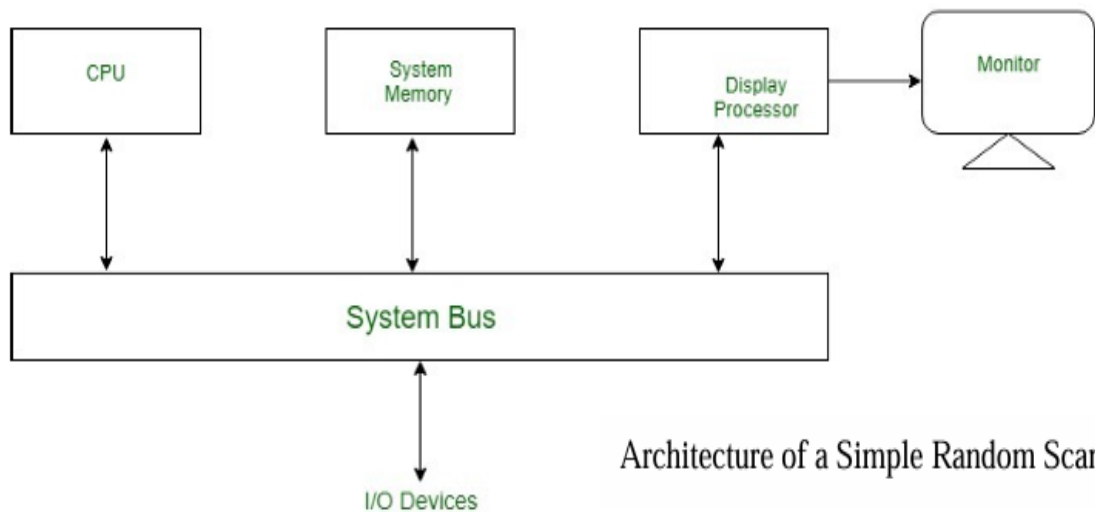


- Calligraphic Displays also called **vector, stroke or line drawing graphics**
- Electron beam directed only to the points of picture to be displayed.
- Picture is stored as a set of point and line-drawing commands with (x,y) or (x,y,z) endpoint coordinates, as well as character-plotting commands.
- Storage referred as **display list, refresh display file, vector file or display program**
- System cycles through set of commands in display list, draw lines and refreshes it.
- Refresh rate depends on the number of lines to be displayed.
- To avoid flicker it must be at least 30 times per second (30 Hz).
- They are designed for smooth line drawing applications, higher resolution.
- Cannot display realistic shaded images.

Random Scan System

Input in the form of an application program is stored in the system memory along with graphics package. Graphics package translates the graphic commands in the application program into a display file stored in system memory. This display file is then accessed by the display processor to refresh the screen. The display processor cycles through each command in the display file

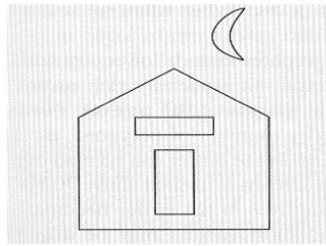
program. Sometimes the display processor in a random scan is referred to as **Display Processing Unit / Graphics Controller**.



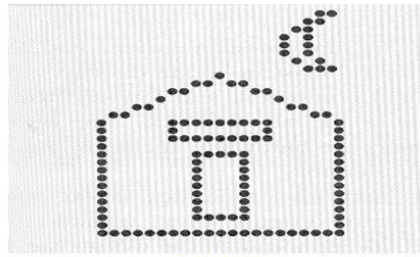
Architecture of a Simple Random Scan System

- Graphics patterns are drawn on a random-scan system by directing the electron beam along the component lines of the picture.
- Lines are defined by the values for their coordinate endpoints,
- these input coordinate values are converted to x and y deflection voltages.
- A scene is then drawn one line at a time by positioning the beam to fill in the line between specified endpoints.
- Graphics patterns are drawn on a random-scan system by directing the electron beam along the component lines of the picture.
- Lines are defined by the values for their coordinate endpoints,
- these input coordinate values are converted to x and y deflection voltages.
- A scene is then drawn one line at a time by positioning the beam to fill in the line between specified endpoints.

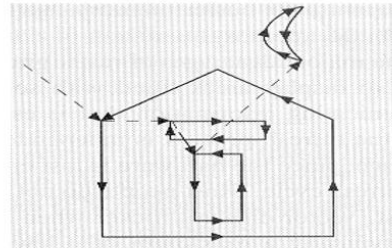
Raster Scan Vs Random Scan



Ideal line drawing

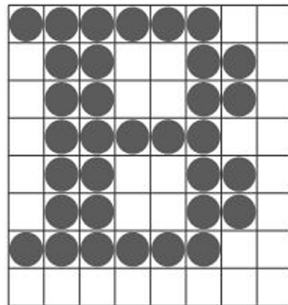


Raster scan



Random scan

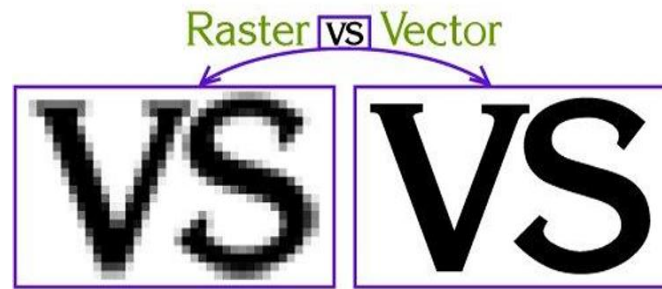
Raster-Scan vs Random-Scan



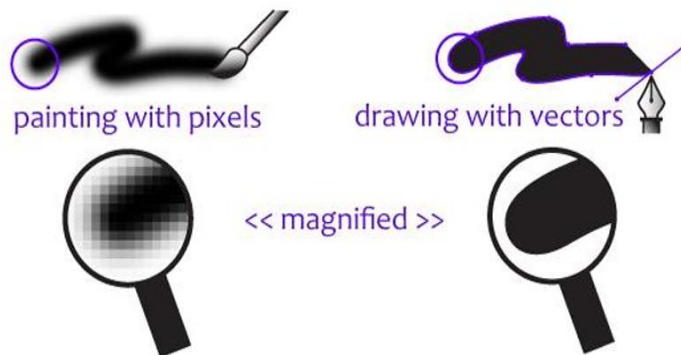
A character defined as a rectangular grid of pixel positions.



A character defined as an outline shape.



Raster (bitmap) images are made of pixels. A pixel is a single point or the smallest single element in a display device.



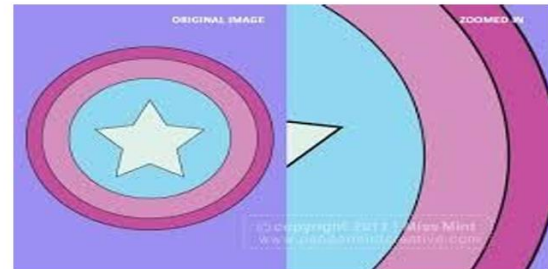
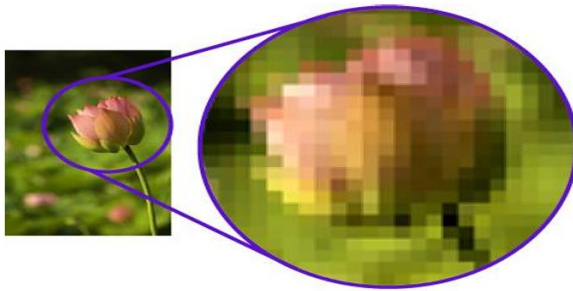
Vector images are mathematical calculations from one point to another that form geometrical shapes.

RASTER SCAN	RANDOM SCAN
While the resolution of raster scan is lesser or lower than random scan	The resolution of random scan is higher than raster scan
While the cost of raster scan is lesser than random scan.	It is costlier than raster scan.
It stores picture definition as a set of intensity values of the pixels in the frame buffer.	It stores picture definition as a set of line commands in the memory.
While in raster scan, any alteration is not so easy.	In a random scan, any alteration is easy in comparison of a raster scan.
The refresh rate is 60 to 80 frames per second and is independent of picture complexity.	The refresh rate depends on the number of lines to be displayed i.e. 30 to 60 times per second.
Eg: TV Sets	Eg : Pen Plotter



When you take a photograph using a digital camera or scan an image from a magazine, you are creating a raster (bitmap) graphic.

Vector graphics are scalable - ie when you resize them, they do not lose quality.



Common file formats:

Raster/Bitmap

bmp (BitMaP)
jpeg (Joint Photographic Experts Group)
Gif (Graphics Interchange Format)
Tiff (Tagged Image File Format)
png (Portable Network Graphics)

Vector

cgm (Computer Graphics Metafile)
svg (Scalable Vector Graphics)
odg (OpenDocument Graphic)
eps (Encapsulated PostScript)
dxf (drawing exchange format)

4) Color CRT Monitors:

A CRT monitor displays color pictures by using a combination of phosphors that emit different colored light. It produces range of colors by combining the light emitted by different phosphors. There are two basic techniques for color display:

1. Beam-penetration technique
2. Shadow-mask technique

1) Beam-penetration technique

- This technique is used with random scan monitors.
- In this technique inside of CRT coated with two phosphor layers usually red and green. The outer layer of red and inner layer of green phosphor.
- The color depends on how far the electron beam penetrates the phosphor layer.

- A beam of fast electron penetrates more and excites inner green layer while slow electron excites outer red layer.
- At intermediate beam speed we can produce combination of red and green lights which emit additional two colors orange and yellow.
- The beam acceleration voltage controls the speed of the electrons and hence color of pixel.
- It is a low-cost technique to produce color in random scan monitors.
- It can display only four colors.
- Quality of picture is not good compared to other techniques.

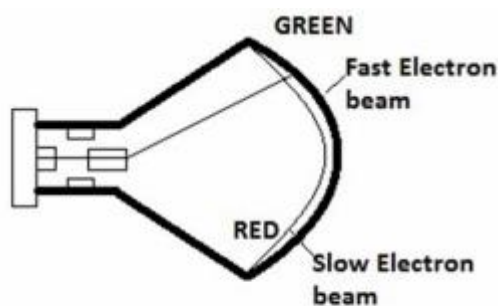


Fig. 1.5: - Beam-penetration CRT

2) Shadow-mask technique

- A shadow mask tube uses a metal plate with tiny holes, placed so that the electron beam only illuminates the correct phosphors on the face of the tube.
- It produces a wide range of colors as compared to beam-penetration technique.
- This technique is generally used in raster scan displays. Including color TV.
- In this technique CRT has three phosphor color dots at each pixel position. One dot for red, one for green and one for blue light. This is commonly known as **Dot Triangle**.
- Here in CRT there are three electron guns present, one for each color dot. And a shadow mask grid just behind the phosphor coated screen.
- The shadow mask grid consists of a series of holes aligned with the phosphor dot pattern.
- Three electron beams are deflected and focused as a group onto the shadow mask and when they pass through a hole they excite a dot triangle.
- In dot triangle three phosphor dots are arranged so that each electron beam can activate only its corresponding color dot when it passes through the shadow mask.

- A dot triangle when activated appears as a small dot on the screen which has color of combination of three small dots in the dot triangle.
- By changing the intensity of the three electron beams we can obtain different colors in the shadow mask CRT.

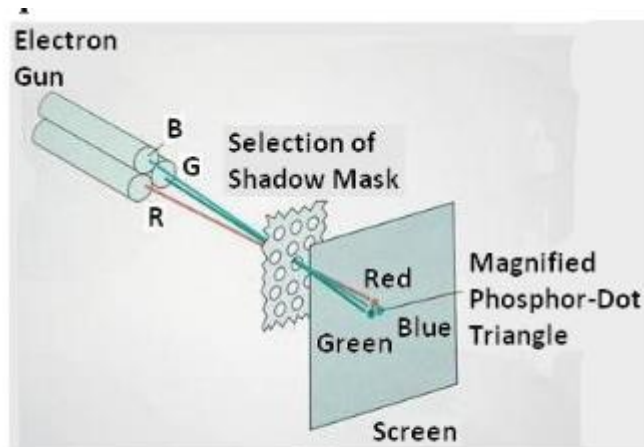
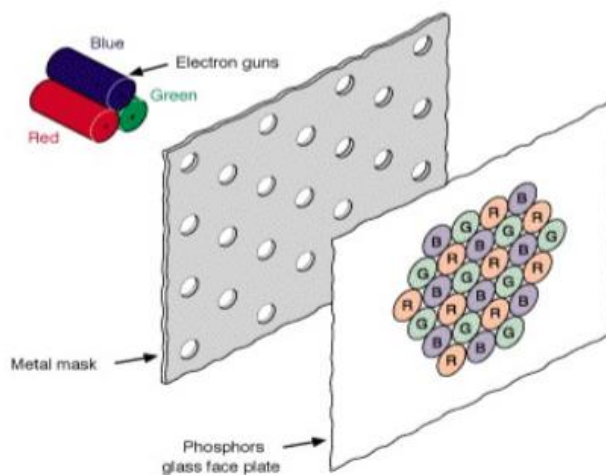
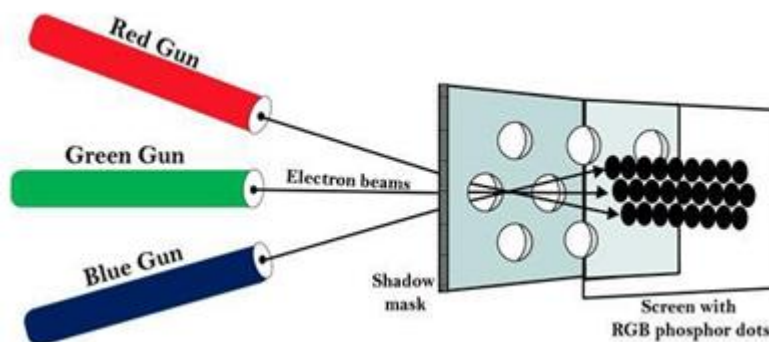


Fig. 1.6: - Shadow-mask CRT.



Delta-Delta Shadow Mask Method



Commonly used in color CRT systems.

The three electron beams are deflected and focused as a group onto the shadow mask, which contains a series of holes aligned with the phosphor dot patterns. When the three beams pass through a hole in the shadow mask, they activate a dot triangle, which appears as a small color spot on the screen. The phosphor dots in the triangles are arranged so that each electron beam can activate only its corresponding color dot when it passes through the shadow mask.

We obtain color variations in a shadow mask CRT by varying the intensity levels of the three electron beams. By turning off the red and green guns, we get only the color coming from the blue phosphor.

Other combinations of beam intensities produce a small light spot for each pixel position since our eyes tend to merge the three colors into one composite.

The color we see depends on the amount of excitation of the red, green, and blue phosphors.

A white area is a result of activating all three dots with equal intensity.

Yellow is produced with the green and red dots only; magenta is produced with the blue and red dots, and cyan shows up when blue and green are activated equally.

Direct-View Storage Tubes

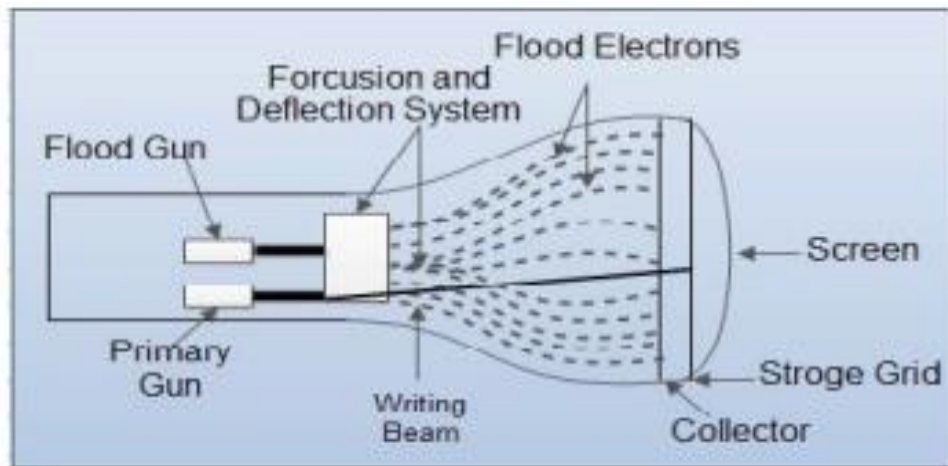
An alternative method for maintaining a screen image is to store the picture information inside the CRT instead of refreshing the screen.

A direct-view storage tube (DVST) stores the picture information as a charge distribution just behind the phosphor-coated screen.

Two electron guns are used in a DVST.

1. primary gun
2. flood gun

The Primary gun used to store the picture as pattern of positive charges on storage grid. The Flood gun used to maintain the picture display. Picture pattern is transferred to phosphor by continuous flood of electron generated from flood gun. Electrons are attracted by positively charged picture pattern on storage grid and are repelled by rest. The attracted electrons strike on phosphor making it visible on screen.



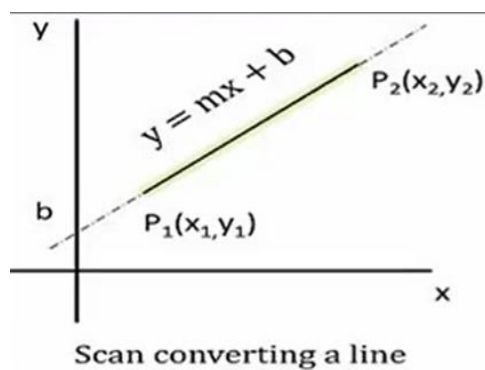
Advantages over refresh CRT

- Very complex pictures can be displayed at very high resolutions without flicker.

Disadvantages of DVST systems

- They do not display colour.
- The erasing and redrawing process can take several seconds for a complex picture.

Line Drawing Algorithms



The **slop Intercept equation** for a straight line is $y = m \cdot x + b$

where m is slope of the line and b is the y -intercept.

- Given that two endpoints of a line segment are specified at positions (x_1, y_1) and (x_2, y_2) then
- $m = \frac{y_2 - y_1}{x_2 - x_1}$ (eq1) ; $b = y_1 - m \cdot x_1$ (eq2)
- another representation $\rightarrow m = \frac{dy}{dx}$
- where $dy = y_2 - y_1$ and $dx = x_2 - x_1$

- $m=dx/dy$
- where $dy=y_2-y_1$ and $dx=x_2-x_1$

$$dy=m.dx$$

$$dx=dy/m$$

Digital Differential Analyzer (DDA) Line Drawing Algorithm

DDA Algorithm is the simplest line drawing algorithm

Case 1: If slope (m) is less than or equal to 1, we sample at unit x intervals ($dx=1$) and

compute each successive y value as $y_{k+1} = y_k + m$ (eq3)

Case 2: If the slope (m) is greater than 1, we sample at unit y intervals ($dy=1$) and compute each successive x value as $x_{k+1} = x_k + 1/m$ (eq4)

Note :

Equations 3 and 4 are based on the assumption that lines are to be **processed from the left endpoints to the right endpoint**. If this processing is reserved ie from the right endpoint to the left endpoint, then equations 3 and 4 become

$$y_{k+1} = y_k - m \quad \& \quad x_{k+1} = x_k - 1/m$$

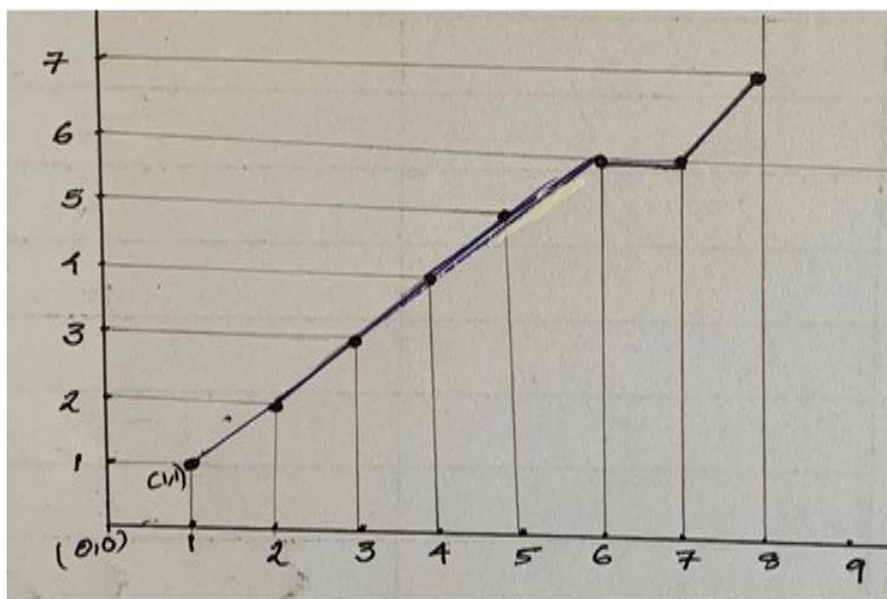
Example 1:

Draw line from (1,1) to(8,7) using the DDA algorithm

$m=(7-1)/(8-1) = 6/7 = 0.9 < 1$ then follow case 1

increment $x_{k+1}=x_k+1$ and $y_{k+1}=y_k+m$ initially plot (1,1)

No	X_{k+1}	Y_{k+1}	Pixel Plotted Round(X_{k+1} , Y_{k+1})
1	$1+1=2$	$1+6/7=1.9$	Round(2,1.9)=(2,2)
2	$2+1=3$	$1.9+6/7=2.8$	Round(3,2.8)=(3,3)
3	$3+1=4$	$2.8+6/7=3.7$	Round(4, 3.7)=(4,4)
4	$4+1=5$	$3.7+6/7=4.6$	Round(5, 4.6)=(5,5)
5	$5+1=6$	$4.6+6/7=5.5$	Round(6, 5.5)=(6,6)
6	$6+1=7$	$5.5+6/7=6.4$	Round(7, 6.4)=(7,6)
7	$7+1=8$	$6.4+6/7=7.3$	Round(8, 7.3)=(8,7)

**Example 2:**

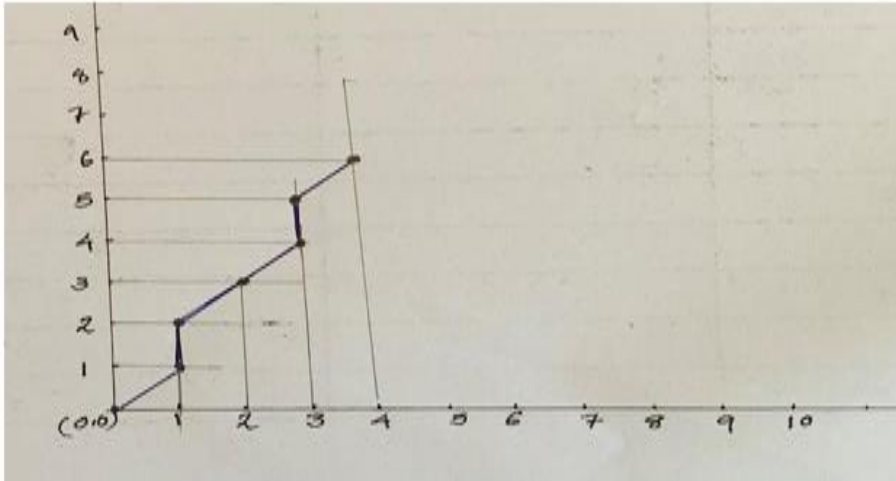
Draw line from (0,0) to(4,6) using the DDA algorithm

$$m = (6-0)/(4-0) = 6/4 = 1.5 > 1 \text{ then follow case 2}$$

increment $y_{k+1} = y_k + 1$ and $x_{k+1} = x_k + 1/m$

initially plot (0,0)

No	X_{k+1}	Y_{k+1}	Pixel Plotted Round(X_{k+1} , Y_{k+1})
1	$0 + 4/6 = 0.67$	$0 + 1 = 1$	Round(0.67,1)=(1 ,1)
2	$0.67 + 0.67 = 1.34$	$1 + 1 = 2$	Round(1.34,2)=(1,2)
3	$1.34 + 0.67 = 2.01$	$2 + 1 = 3$	Round(2.01,3)=(2,3)
4	$2.01 + 0.67 = 2.68$	$3 + 1 = 4$	Round(2.68,4)=(3,4)
5	$2.68 + 0.67 = 3.35$	$4 + 1 = 5$	Round(3.35,5)=(3,5)
6	$3.35 + 0.67 = 4.02$	$5 + 1 = 6$	Round(4.02,6)=(4,6)

**Example 3:-**

If this processing is reserved ie from the right endpoint to the left endpoint, then equations 3 and 4 become

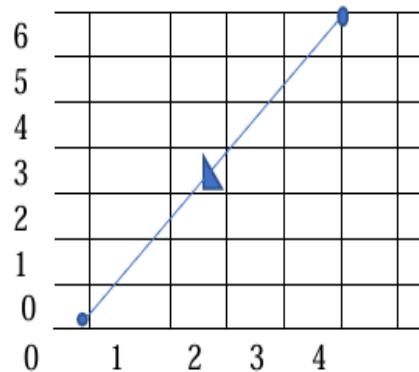
$$y_{k+1} = y_k - m \quad \& \quad x_{k+1} = x_k - 1/m$$

Draw line from (4,6) to (0,0) using the DDA algorithm

$$m = (0-6)/(0-4) = -6/-4 = 1.5 > 1 \text{ then follow case 2}$$

increment $y_{k+1} = y_k - 1$ and $x_{k+1} = x_k - 1/m$

initially plot (4,6)



0	X_{k+1}	Y_{k+1}	Pixel Plotted Round(X_{k+1} , Y_{k+1})
1	3.334	5	Round(3.334,5)=(3 ,5)
2	2.668	4	Round(2.668,4)=(3,4)
3	2.002	3	Round(2.002,3)=(2,3)
4	1.336	2	Round(1.336,2)=(1,2)
5	0.67	1	Round(0.67,1)=(1,1)
6	0.004	0	Round(0.004,0)=(0,0)

Algorithm

```

#define Round(a) ((int)(a+0.5))

void lineDDA(int xa, int ya, int xb, int yb)
{ int dx=xb-xa, dy=yb-ya , steps, k;
  float xIncrement , yIncrement , x=xa , y=ya;

  if(abs(dx)>abs(dy))
    steps=abs(dx);
  else
    steps=abs(dy);

  xIncrement=dx/(float)steps;
  yIncrement=dy/(float)steps;

  setPixel(Round(x) , Round(y));
  for(k=0;k<steps;k++)

    { x+=xIncrement;
      y+=yIncrement;
      setPixel(Round(x),Round(y));
    }
}

```

advantages of DDA Algorithm

- It is a simple algorithm.
- It is easy to implement.
- It avoids using the multiplication operation which is costly in terms of time complexity.

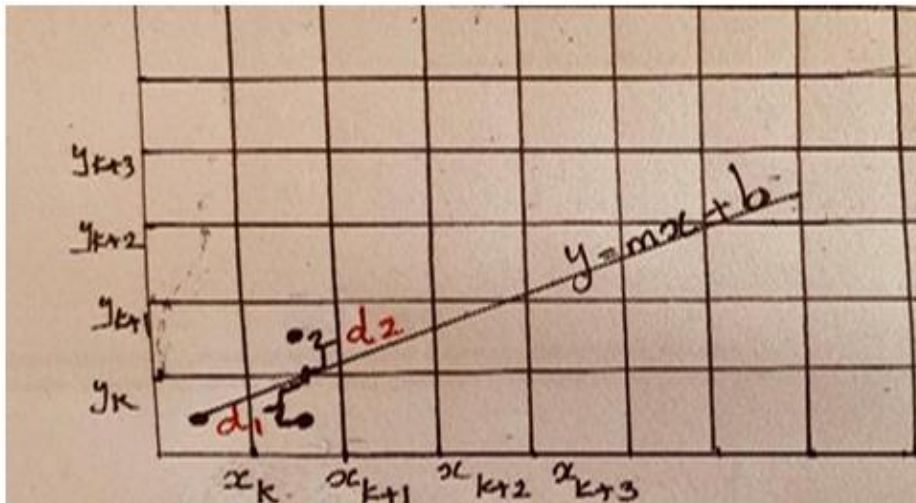
disadvantages of the DDA Algorithm

- There is an extra overhead of using the round-off () function.

- Using the round-off () function increases the time complexity of the algorithm.
- Resulted lines are not smooth because of the round-off () function.
- The points generated by this algorithm are not accurate.

Bresenham's Line Drawing Algorithm

An accurate and efficient raster line generating algorithm.



To illustrate Bresenham's approach, we- first consider the scan-conversion process for lines with positive slope less than 1. Pixel positions along a line path are then determined by sampling at unit x intervals. Starting from the left endpoint (x_0, y_0) of a given line, we step to each successive column (x position) and plot the pixel whose scan-line y value is closest to the line path.

To determine the pixel (x_k, y_k) is to be displayed, next to decide which pixel to plot the column $x_{k+1} = x_k + 1$. At sampling position x_{k+1} , we label vertical pixel separations from the mathematical line path as d_1 and d_2 . The y coordinate on the mathematical line at pixel column position x_{k+1} is calculated as

$$\text{Line equation } y = m \cdot x + b \dots\dots\dots(1)$$

The y coordinate on the mathematical line at pixel column position $x_k + 1$ is calculated as

$$y = m(x_k + 1) + b \dots\dots\dots(2)$$

$$\text{then } d1 = y - y_k$$

$$= m(x_k + 1) + b - y_k \dots\dots\dots(3)$$

$$d2 = (y_k + 1) - y$$

$$= y_k + 1 - m(x_k + 1) - b \dots\dots\dots(4)$$

To determine which of the two pixel is closest to the line path, efficient test that is based on the difference between the two-pixel separations

The difference between $d1$ and $d2$ is

$$\begin{aligned} d1 - d2 &= m(x_k + 1) + b - y_k - (y_k + 1 - m(x_k + 1) - b) \\ &= 2m(x_k + 1) - 2y_k + 2b - 1 \dots\dots\dots(5) \end{aligned}$$

Substitute $m = dy/dx$ in equ 5

$$(d1 - d2)dx = 2 \cdot dy(x_k + 1) - 2y_k \cdot dx + 2b \cdot dx - dx$$

$$(d1 - d2)dx = 2 \cdot dy \cdot x_k + 2 \cdot dy - 2y_k \cdot dx + 2b \cdot dx - dx$$

$$P_k = 2 \cdot dy \cdot x_k - 2dx \cdot y_k + 2 \cdot dy + 2b \cdot dx - dx$$

$$P_k = 2 \cdot dy \cdot x_k - 2dx \cdot y_k + C \dots\dots\dots(6) \quad \text{where } P_k \text{ is decision parameter}$$

if $(P_k \leq 0)$ ie $d1 - d2 < 0$, meaning actual line lays closer to y_k , then next pixel coordinate is $(x_k + 1, y_k)$.

if $(P_k > 0)$ ie $d1 - d2 > 0$, meaning actual line lays closer to $y_k + 1$, then next pixel coordinate is $(x_k + 1, y_k + 1)$.

Calculate successive decision parameters

At step $k+1$ the decision parameter is $p_{k+1} = 2 \cdot dy \cdot x_{k+1} - 2 \cdot dx \cdot y_{k+1} + C \dots\dots\dots(8)$

equ(8)-equ(6)

$$\begin{aligned}
 p_{k+1} - p_k &= 2 \cdot dy \cdot x_{k+1} - 2 \cdot dx \cdot y_{k+1} + C - (2 \cdot dy \cdot x_k - 2 \cdot dx \cdot y_k + C) \\
 &= 2 \cdot dy (x_{k+1} - x_k) - 2 \cdot dx (y_{k+1} - y_k) \\
 \mathbf{p_{k+1}} &= \mathbf{p_k + 2 \cdot dy (x_{k+1} - x_k) - 2 \cdot dx (y_{k+1} - y_k) \dots\dots\dots(9)}
 \end{aligned}$$

The first decision parameter P_0 is evaluated at starting pixel position (x_0, y_0) and with m evaluated as dy/dx , using equation (6)

$$p_k = 2 \cdot dy \cdot x_k - 2 \cdot dx \cdot y_k + 2 \cdot dy + 2 \cdot b \cdot dx - dx$$

$y = mx + b$, $b = y - mx \rightarrow b = y - (dy/dx) \cdot x$ substitute in above equation

$$p_0 = 2 \cdot dy \cdot x_0 - 2 \cdot dx \cdot y_0 + 2 \cdot dy + 2 \cdot (y_0 - (dy/dx) \cdot x_0) \cdot dx - dx$$

$$p_0 = 2 \cdot dy \cdot x_0 - 2 \cdot dx \cdot y_0 + 2 \cdot dy + 2 \cdot y_0 \cdot dx - 2 \cdot dy \cdot x_0 - dx$$

$$\mathbf{p_0 = 2 \cdot dy - dx \dots\dots\dots(12)}$$

Bresenham's Line-Drawing Algorithm for $|m| < 1$

1. Input the two endpoints and store the left endpoint in (x_0, y_0) .
2. Load (x_0, y_0) into the frame buffer; that is plot the first point.
3. Calculate the constants dx , dy , $2dy$, and $2dy-2dx$ and get the first value for the decision parameter as

$$p_0 = 2dy - dx$$
4. At each X_k along the line, starting at $k = 0$, perform the following test –
 If $p_k < 0$, the next point to plot is (x_{k+1}, y_k) and $P_{k+1} = P_k + 2dy$
 Otherwise, the next point to plot is (x_{k+1}, y_{k+1}) and $P_{k+1} = P_k + 2dy - 2dx$
5. Repeat step 4 dx times.

Example: Consider the line with endpoints $(20, 10)$ to $(30, 18)$. Perform Bresenham's Algorithm for line drawing.

Ans: - The line has the slope

$$m = (18-10)/(30-20) = 8/10 = 0.8$$

$$dx = 30-20 = 10$$

$$dy = 18-10 = 8$$

The initial decision parameter has the value

$$p_0 = 2dy - dx = (2*8) - 10 = 6$$

increments for calculating successive decision parameters are

$$2dy = 2*8 = 16 \text{ and } 2dy - 2dx = (2*8) - (2*10) = 16 - 20 = -4$$

We plot the initial point $(x_0, y_0) = (20, 10)$ and determine successive pixel positions along the line path from the decision parameter as

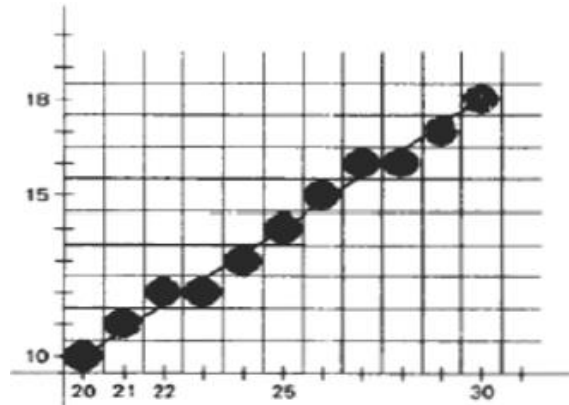
Calculate the constants dx , dy , $2dy$, and $2dy-2dx$ and get the first value for the decision parameter as

$$p_0 = 2dy - dx$$

If $P_k < 0$, the next point to plot is (x_{k+1}, y_k) and $P_{k+1} = P_k + 2dy$

If $P_k > 0$, the next point to plot is (x_{k+1}, y_{k+1}) and $P_{k+1} = P_k + 2dy - 2dx$

k	p_k	(x_k+1, y_k+1)
0	6	(21,11)
1	2	(22,12)
2	-2	(23,12)
3	14	(24,13)
4	10	(25,14)
5	6	(26,15)
6	2	(27,16)
7	-2	(28,16)
8	14	(29,17)
9	10	(30,18)



Advantages

Algorithm is Fast

Uses only integer calculations

Disadvantages

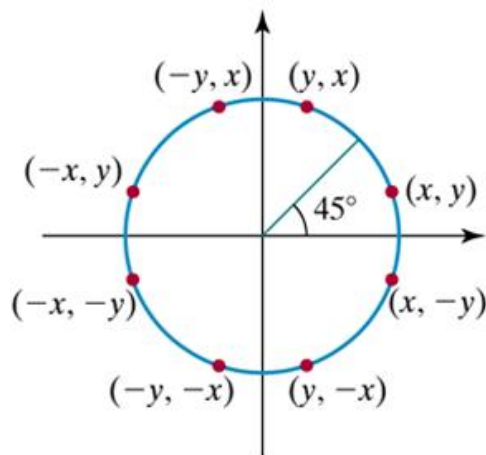
It is meant only for basic line drawing.

Midpoint circle Algorithm:

The mid-point circle drawing algorithm is an algorithm used to determine the points needed for rasterizing a circle.

We use the mid-point algorithm to calculate all the perimeter points of the circle in the first octant and then print them along with their mirror points in the other octants. This will work because a circle is symmetric about its centre.

Consider 8-way circle symmetry and find all points in a single octant and corresponding to each point find out all other seven points of other octants, so after covering a single octant we will get all circle points.



Symmetry of a circle

In the raster line algorithm at unit intervals and determine the closest pixel position to the specified circle path at each step. For a given radius r and screen centre position (x_c, y_c) the pixel positions around a circle path centered at the coordinate position $(0, 0)$.

The circle function as

$$f_{\text{circle}}(x, y) = x^2 + y^2 - r^2$$

Any point (x, y) on the boundary of the circle with radius r satisfies the equation

$$f_{\text{circle}}(x, y) = 0.$$

If the point is in the interior of the circle, the circle function is negative.

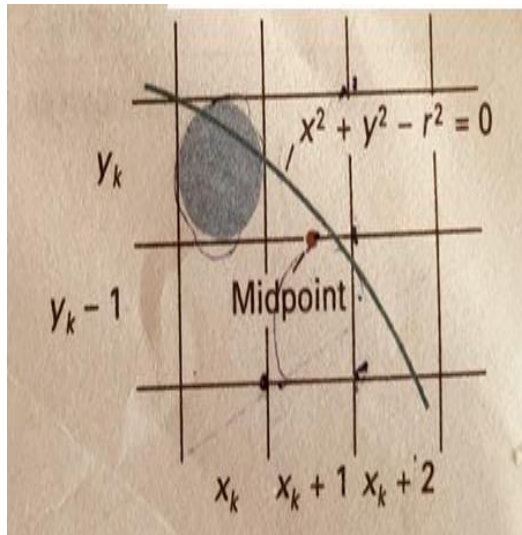
And if the point is outside the circle the circle function is positive.

$$f_{\text{circle}}(x, y) < 0, \quad \text{if } (x, y) \text{ is inside the circle boundary}$$

$$= 0, \quad \text{if } (x, y) \text{ is on the circle boundary}$$

$$> 0, \quad \text{if } (x, y) \text{ is outside the circle boundary}$$

The tests in the above equation are performed for the mid positions between pixels near the circle path at each sampling step. The circle function is the decision parameter in the midpoint algorithm.



Decision making Assuming we have plotted the pixel at (x_k, y_k) , we need to determine the next pixel at (x_{k+1}, y_{k+1}) . Choose one pixel from $(x_k + 1, y_k)$, (x_{k+1}, y_{k-1}) which is closest to the circle boundary. Use circle function $f_{\text{circle}}(x, y) = x^2 + y^2 - r^2$ as the decision parameter and take midpoint of (x_{k+1}, y_k) and (x_{k+1}, y_{k-1}) as its x and y values.

$$P_k = f_{\text{circle}}(x_{k+1}, y_{k-1/2}) \\ = (x_{k+1})^2 + (y_{k-1/2})^2 - r^2 \dots (1)$$

Midpoint of x_k+1, x_{k+1} $(x_{k+1} + x_{k+1})/2 = (2 \cdot x_{k+1})/2$ $= x_{k+1}$	Midpoint of y_k, y_{k-1} $(y_k + y_{k-1})/2 = (2y_{k-1})/2$ $= y_{k-1/2}$
--	---

If $p_k < 0$, this midpoint is inside the circle and the pixel on the scanline y_k is closer to the circle the boundary, and we select y_k

Otherwise, the midpoint is outside or on the circle boundary and we select the pixel on scan line y_{k-1} .

Successive decision parameters are obtained using incremental calculations.

We obtain a recursive expression for the next decision parameter by evaluating the circle function at the sampling position $(x_{k+1}+1, y_{k+1}-1/2)$

$$\begin{aligned}
 P_{k+1} &= (x_{k+1}+1)^2 + (y_{k+1}-1/2)^2 - r^2 \\
 &= (x_k+1+1)^2 + (y_{k+1}-1/2)^2 - r^2 \\
 &= (x_k+2)^2 + (y_{k+1}-1/2)^2 - r^2 \\
 &= x_k^2 + 4x_k + 4 + (y_{k+1})^2 - (y_{k+1}) + 1/4 - r^2
 \end{aligned}$$

$$\begin{aligned}
 P_k &= (x_k+1)^2 + (y_k-1/2)^2 - r^2 \\
 &= x_k^2 + 2x_k + 1 + (y_k)^2 - (y_k) + 1/4 - r^2
 \end{aligned}$$

$$\begin{aligned}
 P_{k+1} - P_k &= x_k^2 + 4x_k + 4 + (y_{k+1})^2 - (y_{k+1}) + 1/4 - r^2 - (x_k^2 + 2x_k + 1 + (y_k)^2 - 2(y_k) + 1/4 - r^2) \\
 &= 2x_k + 3 + (y_{k+1})^2 - (y_{k+1}) - (y_k)^2 + (y_k)
 \end{aligned}$$

$$P_{k+1} = P_k + 2x_k + 3 + (y_{k+1})^2 - (y_{k+1}) - (y_k)^2 + (y_k)$$

If $p_k < 0$, then point (x_k+1, y_k)

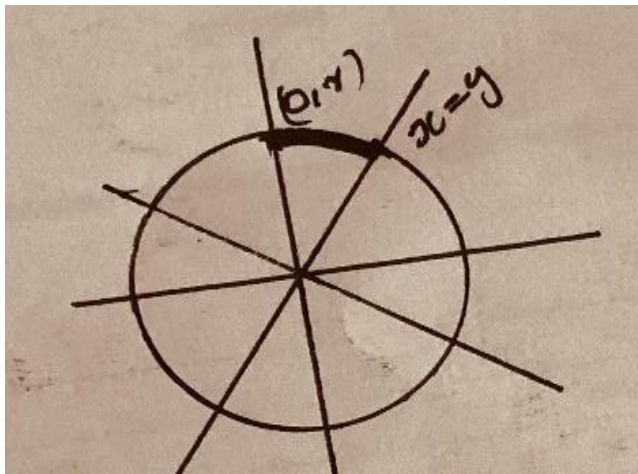
$$\begin{aligned}
 P_{k+1} &= P_k + 2x_k + 3 + y_k^2 - y_k - y_k^2 + y_k \\
 &= P_k + 2x_k + 3
 \end{aligned}$$

If $p_k > 0$, then point (x_k+1, y_k-1)

$$\begin{aligned}
 P_{k+1} &= P_k + 2x_k + 3 + (y_{k-1})^2 - (y_{k-1}) - (y_k)^2 + (y_k) \\
 &= P_k + 2x_k + 3 + y_k^2 - 2y_k + 1 - y_k^2 + y_k \\
 &= P_k + 2x_k - 2y_k + 5
 \end{aligned}$$

Initial Decision parameter is obtained by evaluating the circle function at the position

$(x_0, y_0) = (0, r)$



$$P_0 = f_{\text{circle}}(1, r-1/2) \\ = 1 + (r-1/2)^2 - r^2$$

$$P_0 = 5/4 - r,$$

if the radius r is specified as an integer, we can simply round p_0 to
 $p_0 = 1 - r$.

Midpoint Circle Algorithm

1. Input radius r and circle center (x_0, y_0) , and obtain the first point on the circumference of a circle centered on the origin as

$$(x_0, y_0) = (0, r)$$

2. Calculate the initial value of the decision parameter as $P_0 = 5/4 - r$
3. At each x_k position, starting at $k=0$, perform the following test: if $p_k < 0$, the next point along the circle centered on $(0,0)$ is (x_{k+1}, y_k) and

$$P_{k+1} = P_k + 2x_k + 3$$

Otherwise next point along the circle is (x_{k+1}, y_{k-1}) and

$$P_{k+1} = P_k + 2x_k - 2y_k + 5$$

4. Determine symmetry points in the other seven octants.
5. Move each calculated pixel position (x, y) onto the circular path centered on (x_c, y_c) and plot the coordinate values:

$$x = x + x_c \quad y = y + y_c$$

Example

Given a circle radius $r=10$, determine positions along the circle octant in the first quadrant from $x=0$ to $x=y$.

The initial value of the decision parameter is

$$P_0 = 1 - r = 1 - 10 = -9$$

Initial point is $(x_0, y_0) = (0, 10)$

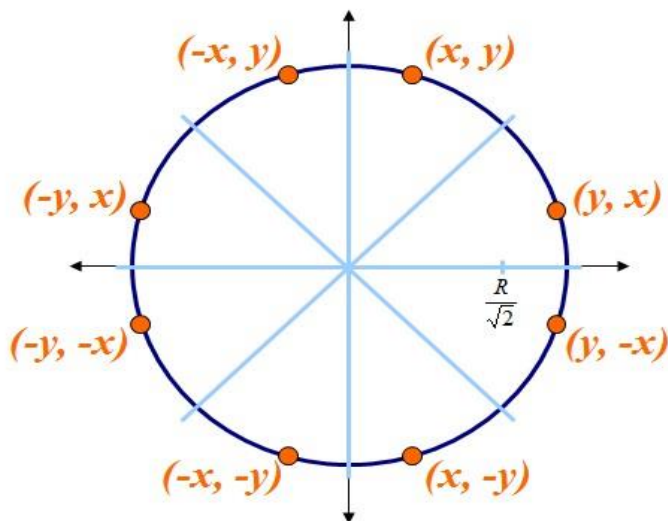
$$2x_0=0 \quad 2y_0=20$$

k	pk	(x_{k+1}, y_{k+1})	$2x_k+3$	$2x_k-2y_k+5$
0	-9	(1,10)	3	
1	$-9+3=-6$	(2,10)	5	
2	$-6+5=-1$	(3,10)	7	
3	$-1+7=6$	(4, 9)		$6-20+5=-9$
4	$6-9=-3$	(5,9)	11	
5	$-3+11=8$	(6, 8)		$10-18+5=-3$
6	$8-3=5$	(7,7) $\rightarrow x=y$,then stop		

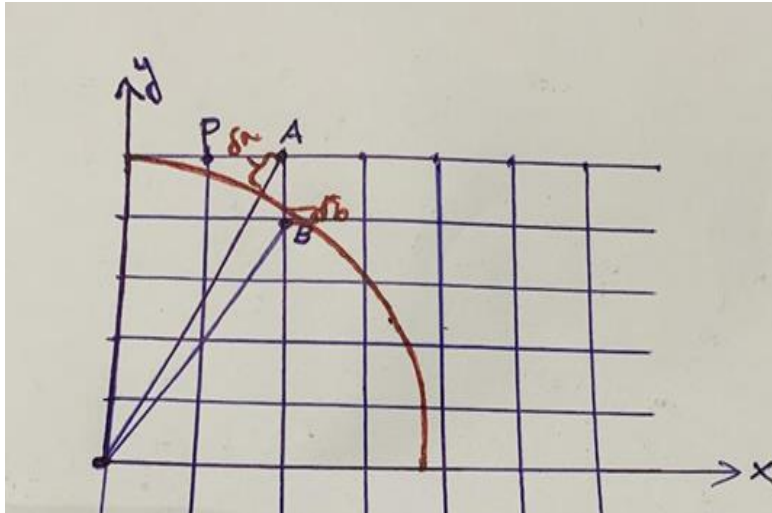
Corresponding to each obtained point, find out 7 other symmetric points also in other 7 octants and draw the complete circle.

Bresenham's circle drawing algorithm

Circles have the property of being highly symmetrical, which is handy when it comes to drawing them on a display screen.



- We know that there are 360 degrees in a circle. First, we see that a circle is symmetrical about the x axis, so only the first 180 degrees need to be calculated.
- Next, we see that it's also symmetrical about the y axis, so now we only need to calculate the first 90 degrees. Finally, we see that the circle is also symmetrical about the 45-degree diagonal axis, so we only need to calculate the first 45 degrees.



Distance of pixel A from the origin (0,0)

$$D_A = \sqrt{(x_i + 1)^2 + y_i^2}$$

Distance of pixel B from the origin (0,0)

$$D_B = \sqrt{(x_i + 1)^2 + (y_i - 1)^2}$$

The distance of pixel A and B from the true circle are

$$d_a = D_A - r$$

$$d_b = D_B - r$$

To avoid square root take a square

$$d_a = D_A^2 - r^2$$

$$d_b = D_B^2 - r^2$$

$$d_a = (x_i + 1)^2 + y_i^2 - r^2$$

$$d_b = (x_i + 1)^2 + (y_i - 1)^2 - r^2$$

Decision Variable **$d_i = d_a + d_b$**

$$d_i = 2(x_i + 1)^2 + 2y_i^2 - 2y_i - 2r^2 + 1$$

Successive Decision parameter

$$d_{i+1} = 2(x_{i+1} + 1)^2 + 2y_{i+1}^2 - 2y_{i+1} - 2r^2 + 1$$

$$= 2(x_i + 2)^2 + 2y_{i+1}^2 - 2y_{i+1} - 2r^2 + 1$$

$$d_{i+1} - d_i = 2(x_i + 2)^2 + 2y_{i+1}^2 - 2y_{i+1} - 2r^2 + 1 - (2(x_i + 1)^2 + 2y_i^2 - 2y_i - 2r^2 + 1)$$

$$d_{i+1} = d_i + 4x_i + 2y_{i+1}^2 - 2y_{i+1} - 2y_i^2 + 2y_i + 6$$

Case1: $d_i < 0$

$$x_{i+1} = x_i + 1$$

$$y_{i+1} = y_i$$

$$d_{i+1} = d_i + 4x_i + 2y_i^2 - 2y_i - 2y_i^2 + 2y_i + 6;$$

$$d_{i+1} = d_i + 4x_i + 6$$

Case2: $d_i \geq 0$

$$x_{i+1} = x_i + 1$$

$$y_{i+1} = y_i - 1$$

$$d_{i+1} = d_i + 4x_i + 2(y_i - 1)^2 - 2(y_i - 1) - 2y_i^2 + 2y_i + 6$$

$$d_{i+1} = d_i + 4x_i + 2y_i^2 - 2y_i + 2 - 2y_i + 2 - 2y_i^2 + 2y_i + 6$$

$$d_{i+1} = d_i + 4x_i - 4y_i + 10$$

Initial Decision Parameter

Initial Pixel at (0,r)

$$d_i = 2(x_i + 1)^2 + 2y_i^2 - 2y_i - 2r^2 + 1$$

$$d_i = 2(0 + 1)^2 + 2r^2 - 2r - 2r^2 + 1$$

$$= 2 + 2r^2 - 2r - 2r^2 + 1$$

$$d_0 = 3 - 2r$$

Bresenham's Circle Drawing Algorithm

1. Input radius r and circle center (x_c, y_c) , then set the coordinates for the first point on the circumference of a circle centered on the origin as $(x_0, y_0) = (0, r)$.
2. Calculate the initial decision parameter as $d_0 = 3 - 2r$
3. At each x_i , from $i = 0$ perform the following :
 If $d_i < 0$, next point to plot along the circle centered on (0,0) is (x_{i+1}, y_i) and

$$d_{i+1} = d_i + 4x_i + 6.$$

 Otherwise ,
 Next point to plot is (x_{i+1}, y_{i-1}) and

$$d_{i+1} = d_i + 4(x_i - y_i) + 10.$$
4. Determine the symmetry points in the other seven octants.
5. Move each calculated pixel position (x, y) onto the circular path centered at (x_c, y_c) and plot the coordinate values: $x = x + x_c$; $y = y + y_c$;
6. Repeat steps 3 through 5 until $x \geq y$.

Example

Given a circle radius $r=10$, determine positions along the circle octant in the first quadrant from $x=0$ to $x=y$. The initial value of the decision parameter is

$$d_0 = 3 - 2 \times 10 = -17$$

Initial point is $(x_0, y_0) = (0, 10)$

k	dk	(x_{k+1}, y_{k+1})	$4x_i + 6$	$4(x_i - y_i) + 10$
0	-17	(1,10)	$4 \times 0 + 6 = 6$	
1	$-17 + 6 = -11$	(2,10)	$4 \times 1 + 6 = 10$	
2	$-11 + 10 = -1$	(3,10)	$4 \times 2 + 6 = 14$	
3	$-1 + 14 = 13$	(4,9)		$4(3 - 10) + 10 = -18$
4	$13 - 18 = -5$	(5,9)	$4 \times 4 + 6 = 22$	
5	$-5 + 22 = 17$	(6,8)		$4(5 - 9) + 10 = -6$
6	$17 - 6 = 11$	(7,7) $x=y$ then stop		

Problems

1) Suppose you have a raster system designed using an 8 inches x 10 inches screen with a resolution of 100 pixels per inch in each direction. What frame buffer size is required if 6 bits are stored per pixel in the buffer?

Answer: Resolution = 8 inch x 10 inch

Each inch with 100 pixels so total $8 \times 100 \times 10 \times 100 = 800 \times 1000$ pixels.

One pixel can store 6 bits, $800 \times 1000 \times 6 = 4800000$ bits = 600000 bytes (1 byte = 8 bits)

= 6×10^5 bytes.

Note:-

$$4,800,000 \div 8 = 600,000 \text{ bytes}$$

3) How much time is spent scanning across each row of pixels during screen refresh on a raster system with a resolution of 1280×1024 and are the refresh rate of 60 frames per second?

Answer-

That means system contains 1024 scan lines and each scan line contains 1280 pixels.

refresh rate = 60 frame/sec.

Step 1: Total time per frame

Time for one complete frame refresh is: Time per frame = $1/60$ seconds

$$=0.01667 \text{ seconds} \quad =16.67 \text{ ms}$$

Step 2: Time per row

There are 1024 rows, so divide the frame time by number of rows:

$$\text{Time per row} = 16.67 \text{ ms} / 1024$$

$$\approx 0.01628 \text{ ms}$$

$$= 16.28 \mu\text{s} \text{ Time per row}$$

3) Consider a raster system with a resolution of 1024×1024 . Compute the size of the raster needed to store 4 bits per pixel? How much storage is needed if 8 bits per pixel are to be stored?

Answer:-

Resolution: 1024×1024 pixels

Pixels per image: $1024 \times 1024 = 1,048,576$ pixels

Case 1: 4 bits per pixel

► Total bits: $1,048,576 \text{ pixels} \times 4 \text{ bits/pixel}$

$$= 4,194,304 \text{ bits}$$

► Convert to bytes:

$$4,194,304 \div 8 = 524,288 \text{ bytes}$$

► Convert to kilobytes (KB): $524,288 \div 1024$

$$= 512 \text{ KB}$$

4) A raster system has a resolution of 800×600 . Calculate the frame buffer size required to store an image with 24 bits per pixel.

Raster: 800×600 , 24 bits per pixel

- Pixels = $800 \times 600 = 480,000$
- Bits = $480,000 \times 24 = \mathbf{11,520,000 \text{ bits}}$
- Bytes = $11,520,000 \div 8 = \mathbf{1,440,000 \text{ bytes}}$
- KB = $1,440,000 \div 1024 \approx \mathbf{1,406.25 \text{ KB} \approx 1.37 \text{ MB}}$

5) Determine the storage requirement for a monochrome image of resolution 640×480 pixels.

Monochrome image: 640×480

1 bit/pixel

$$\text{Pixels} = 640 \times 480 = 307,200 \text{ Bits}$$

$$= 307,200 \times 1 = 307,200 \text{ bits}$$

$$\text{Bytes} = 307,200 \div 8 = 38,400 \text{ bytes}$$

$$\text{KB} = 38,400 \div 1024 = 37.5 \text{ KB}$$

- 6) Compute the size of a 640 x 480 image at 240 pixels per inch

Answer:

$$\text{Width} = 640 \text{ pixels}$$

$$\text{Height} = 480 \text{ pixels}$$

$$\text{Resolution} = 240 \text{ ppi}$$

$$\text{Width} = 640/240 = 2.67 \text{ inches}$$

$$\text{Height} = 480/240 = 2.02 = 2.0 \text{ inches}$$

The size of the image is 2.67 inches $\times$ 2.0 inches.

- 7) Compute the resolution of a 2 x 2-inch image that has 512 x 512 pixels

$$\text{Answer: } - 512 / 2 = 256 \text{ pixels per inch}$$

- 8) What is an image's aspect ratio?

Answer: - The ratio of its width to its height, measured in unit length or number of pixels

- 9) If an image has a height of 2 inches and an aspect ratio of 1.5, what is its width?

Answer: - width = aspect ratio $\times$ height

$$= 1.5 \times 2 = 3 \text{ inches}$$

- 10) If we want to resize a 1024 X 768 image to one that is 640 pixels wide with the same aspect ratio, what would be the height of the resized image?

Answer: - Height = $\frac{640 \times 768}{1024}$

$$= 480$$